Price Signal

Technical report · ModelX 2.0, Phase 2: Implementation · Domain 01: Cost of living and everyday economics

Author: Nivetha Ramanan (SLIIT) · Repository: github.com/IT25101414/price-signal · Data: Central Bank of Sri Lanka Daily Price Reports, 28 July 2020 to 15 October 2026 (the evaluation uses the table to 13 October) Revised 18 October 2026

1. Summary

The problem. Small food buyers in Colombo cannot tell when a price move at Dambulla, the main wholesale source, is about to reach Pettah or Narahenpita. The Central Bank publishes the prices every working day, but only as a PDF that compares a price with yesterday or last Friday.

What we built. A pipeline that downloads the reports politely, reads every price table from the PDFs, keeps the table current each working day, and estimates for each item how much of the gap between a Colombo price and its Dambulla price usually closes in the next 3 report days. The Flask and SQLite website has six pages: a plain-language Welcome page, the Today command centre, Explore, Weekly report, Method and Help. A Streamlit dashboard provides a second interface. A guard sits in front of the user: when data are old, a newer report is unread, a market was closed, a price is missing or the signal is too small, the page gives the reason instead of a number. The guard runs again on every request, not only when cards are written.

What we found. Adding the Dambulla price improved forecasts of Colombo price changes beyond a model that already uses the price's own history. The 95% interval is above zero in all nine series-and-horizon cases, a placebo shows nothing, and the gain is similar in each test year, with one exception reported in section 8.1. A gradient-boosting challenger, tested under a rule fixed in advance, did not beat the ridge model, so the ridge model stays. The effect is modest. In a simulation, a buyer following the model paid 1.2 to 1.6% less than one who always buys at once, about Rs 60 to 96 per 10 kg. The stated 80% range held 76.1 to 82.3% of outcomes the model had not seen, across the three series at 1, 3 and 5 report days.

What is not done. We have not shown the cards to real buyers, so the claim that a buyer would act on them is untested. The daily refresh runs from a script, not a scheduled cloud job, and the website is not yet hosted publicly. A Sinhala and Tamil version was not built. Section 9 lists every gap.

2. The proposal against what exists

What the proposal promised	Status	Evidence or reason
Downloader that obeys robots.txt and its 10-second delay, with a manifest	Done	`download_cbsl_reports.py`; 2,271 reports; `data/manifest.csv`
Position-based parser that rejects unknown layouts	Done	`extract_prices.py`; 1,487 report days; 125 spot checks
Lead-lag evidence with a placebo and intervals	Done	`robust_check.py`, `leadlag_check.py`, `results_robust.txt`, `results_leadlag.txt`
Rupee backtest of the buying decision	Done	`decision_backtest.py`, `results_decision.txt`
Prediction intervals calibrated on held-out data	Done	`model.py --evaluate`: 77.4 to 81.5% inside the 80% range at 3 report days, 76.1 to 82.3% across 1, 3 and 5 (target 75 to 85%), `results_model*.txt`
Guard: stale, unread report, closed market, missing price, too little data, no clear signal	Done	`guard.py`; public holidays supported through `data/holidays.csv`; `refusal_report.py` runs 12 cases
Daily refresh that fails safely	Done	`refresh.py`; a real source failure was handled (section 6)
Interface with the data date on every card	Done	Website in `web/` (Flask, SQLite) and Streamlit `app.py`
Past-price charts and a weekly comparison	Done	Explore and Weekly report pages; weekly CSV and print to PDF; `data/recent_prices.csv` gives a fresh clone real history
Evidence page generated from the result files	Done	Method page; `web/evidence.py` reads `results_*.txt`, so page and code cannot drift apart
Automated tests on every push	Done	GitHub Actions workflow `tests.yml`; 85 tests locally
GitHub Actions daily schedule	Replaced	`run_daily.sh` instead (the proposal's own cut list, item 3). A cloud schedule would need the price table in the public repository; see section 5.4
Two or three buyer interviews	Done	3 consented sessions; median ratings 4/5
Gradient-boosting challenger	Done, not adopted	`challenger_check.py`, `results_challenger.txt`; it did not beat ridge (section 8.7)
Sinhala and Tamil interface	Not done	Cut list item 2
Written permission from the Central Bank	Received	Reply of 12 October 2026: the data are public and need no approval; attribution is requested (section 5.4)

3. Architecture

Data moves in one direction, and nothing reaches the user without passing the guard. The same guard logic runs again on every request to the website and the dashboard, because the cards file can be older than the page view.

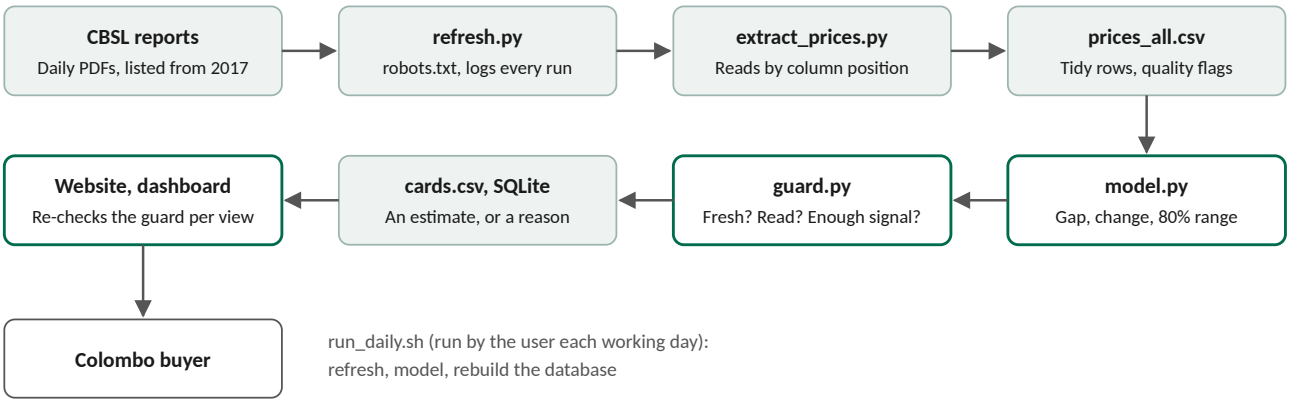


Figure 1. The pipeline. The guard is the only way from the model to the interface.

Part	What it does
`download_cbsl_reports.py`	Reads the index page, obeys robots.txt and its 10-second crawl delay, downloads each PDF once, keeps raw files untouched, writes a manifest (URL, date, SHA-256, time).
`extract_prices.py`	Reads each price table by the horizontal position of every word, because the PDF text layer drops empty cells and splits numbers. Unknown layouts are rejected, never guessed. Failures go to `failed_files.txt`.
`refresh.py`	Reads only the newest listing page, downloads and parses any report not yet in the table, appends it once, and writes one line to `data/refresh_log.csv` with a status: ok, no_new_report, behind, partial, failed or source_unavailable.
`guard.py`	Decides per item whether a number may be shown. First failing check wins: stale (over 3 working days, weekends and listed holidays excluded), report not read, market closed, no Dambulla price, no Colombo price, insufficient data (fewer than 40 of the last 60 report days with both prices), no clear signal.
`model.py`	Ridge regression per Colombo series with a split-conformal 80% range (section 7). Every card passes through the guard. Writes `cards.csv`.
`web/`	Flask server, SQLite database built from `cards.csv` and the price history, and six static pages (Welcome, Today, Explore, Weekly report, Method and Help), with hand-drawn SVG charts, no build step and nothing loaded from other sites. The guard runs on every request: a card that is not ok loses its numbers, and past prices, weekly results and item lists are withheld when the report is stale or a newer one is unread.
`export_recent.py`	Writes `data/recent_prices.csv`, the last 375 days of the four price series shown, so a fresh clone has real charts without the full table.
`challenger_check.py`	Compares a gradient-boosting model with the ridge model on the same split and scoring, under an adoption rule fixed before the run (section 8.7).
`app.py`	Streamlit dashboard with the same rules, kept as a second interface.

4. Technology and why

Part	Choice	Why, and anything that changed from the proposal
Language	Python 3 (3.9 on the development machine)	One language for scraping, parsing, modelling and the interface.
Download	requests, `urllib.robotparser`	Simple, controllable pacing. robots.txt is fetched with our own identifiable User-Agent (section 6).
PDF parsing	pdfplumber	Gives the position of every word, which the parser needs.
Modelling	numpy ridge regression (closed form)	Few parameters, every number explainable. scikit-learn is used only in the early jump-persistence check.
Data handling	pandas, CSV files, SQLite for the website	The proposal said SQLite "if needed". The CSV files stay the source of truth; the website loads them into SQLite so the page can query one item at a time.
Interface	Flask and plain HTML, CSS, JavaScript with inline SVG charts; Streamlit dashboard	Light and dark themes, a table behind every chart, direction shown with arrows and signs as well as colour, and a layout that holds at phone width. Nothing is loaded from other sites.
Scheduling	`run_daily.sh`	Changed. The proposal planned GitHub Actions; see section 5.4. Actions is used for tests instead.
Testing	pytest; GitHub Actions on every push	85 tests (refresh, guard including holidays, model, dashboard, website routes, the challenger rule), plus the extractor's 125 spot checks and `refusal_report.py`.

Everything is free and open source. Nothing costs money. The AI tools used are disclosed in section 12.

5. The data we finally used

5.1 Source and access

Central Bank of Sri Lanka, Statistics Department, Daily Price Report, published at cbsl.gov.lk/en/statistics/economic-indicators/price-report. In the current layout each report is a two-page PDF: page 1 summarises the largest moves, and page 2 is a table of wholesale and retail prices for about 40 items in Pettah, Dambulla, Narahenpita and the fish markets of Peliyagoda and Negombo, with the previous reported price beside each. The prices are by item and market only; nothing identifies a person.

5.2 What we obtained

Measure	Value
Report PDFs downloaded (April 2017 to 13 October 2026; counts in this table are as of that date)	2,271
Reports in the layout the parser reads (from 28 July 2020)	1,495
Report days parsed	1,487 (28 July 2020 to 13 October 2026)
Price rows	268,577, with no duplicate report, item and market rows
Older layouts, rejected on purpose (2017 to mid-2020)	776
Scanned PDFs with no text layer (2022-03-16, 2025-09-24, 2025-09-25, 2026-04-15)	4
Header names a weekday but no date (2026-04-02, 2026-05-29, 2026-07-30)	3
Exact duplicate file (the 22 June 2022 file is a copy of 21 June)	1

5.3 Cleaning, and what was wrong with the source

Layout drift. The report layout changed several times. The parser's header cut-off, column positions and scale adapt to the page, and three bugs found along the way (a header row below a fixed cut-off, large-scale pages, and a weekday-only comparison header) were fixed. Files that return no rows now count as failures instead of passing silently, which exposed 36 empty 2021 files that had been hidden.

Comparison basis. Monday reports compare with last Friday, and some reports compare with a named date. The basis is stored on every row.

Source errors. One report (2 June 2022) quotes a summary figure that disagrees with its own table; we follow the table. About 0.5% of compared prior-versus-today pairs across consecutive reports differ, and are flagged.

Names and values. Item names drift (case, "(lmp)" and "(Imp)"); they are normalised in the analysis. "n.a.", a blank cell and a closed market are kept apart. Prices are rounded to the nearest Rs 5 or 10, so cheap items move in steps.

5.4 Licence, permission and ethics

robots.txt permits automated access at a 10-second crawl delay and blocks only system and search paths (`/admin/`, `/search/`, `/user/` and similar), which we do not touch. We found no statement of reuse terms on the site, so we asked in writing on 11 October 2026. The Statistics Department of the Central Bank replied on 12 October 2026; page 1 of the reply is in the repository (`docs/cbsl_reply_2026-10-12.pdf`, contact details removed). We credit the Statistics Department wherever the data appear, state on the website and in the dashboard that the Central Bank does not endorse the project or its findings, and use the data only for this non-commercial project.

The Central Bank's reply says the Daily Price Report data are public and need no specific approval, that it bears no responsibility for and does not endorse any project built on them, and that the source may be cited as "Statistics Department, Central Bank of Sri Lanka". We follow that wording on every page. We still do not redistribute the raw PDFs. The full price table (`prices_all.csv`) stays local and git-ignored; what the repository holds is a small slice of the published prices (`data/recent_prices.csv`, the last 375 days of the series the website shows) so that a reader can see the charts without a six-hour download. A scheduled cloud refresh would need the full table and a place to keep it, which we have not set up.

6. Temporality

Both historical and live, for different jobs. Years of history train and test the model, because the Dambulla-to-Colombo pattern can only be learned and checked over many months. The latest report drives what the user sees, because a buyer decides today.

Refresh. `refresh.py` checks the newest listing page, adds only reports not yet in the table, and logs every run. A report that cannot be read is logged and left out, and the guard then withholds estimates because the site lists a report the table does not hold. On 13 October the refresh added two reports and the cards moved to the report of that date.

Freshness rules, as built. Every card shows the date of the latest report used. After more than 3 working days without a good report, every number is hidden and the reason is shown (Figure 3). Working days are Monday to Friday minus the dates in `data/holidays.csv`, which lists the 25 distinct 2026 public holiday dates (26 holidays, two on 1 May), taken from a published calendar and to be checked against the government's official calendar. A test shows that a listed holiday stops counting as a working day. The file covers 2026 only and must be extended for later years; a day missing from the file makes good data look stale, and the guard errs toward refusing.

A real failure, handled. On 12 October the first live refresh stopped with `source_unavailable: robots.txt disallows the price page`. We did not skip the check. The cause was our own: the downloader fetched robots.txt with Python's default User-Agent, the server answered that with HTTP 403, and Python's parser treats a 403 as "everything is forbidden". With our identifiable User-Agent the same file returns HTTP 200, and its rules do not restrict the price pages. The archive had downloaded normally on 10 October with the same code, so the server's behaviour towards that User-Agent changed in between. The fix reads robots.txt with our own User-Agent and still stops on any response other than HTTP 200. After the fix the refresh read the index and reported `no_new_report`.

7. Method

Target. The log price change of a Colombo series (wholesale Pettah, retail Pettah, retail Narahenpita) over the next 3 report days; also 1 and 5 in the evaluation.

Model. Ridge regression, pooled across items. Own-history model C uses the series' own 1-day and 5-day changes and its distance from its own 20-day mean. Model D adds the Dambulla price's 1-day and 5-day changes and the Dambulla-minus-Colombo gap measured against its own trailing 60-day average. The gain from Dambulla is D minus C, so it is what Dambulla adds beyond everything the price's own history explains.

Time split. Fit to 2023, test on 2024 onward (for the range check: fit to 2022, calibrate on 2023, test on 2024 onward). Labels that reach past the cut-off are dropped. Intervals use whole months resampled 2,000 times.

Range. Split-conformal. Residuals are measured on a calibration year the model was not fitted on, scaled by each item's recent volatility, and the 80% quantile gives the range. In the live run the model is fitted up to a year before the latest report and calibrated on the last year.

Signal labels. The weak cut-off is the median absolute prediction on the training period, so about half of cards read "no clear signal" by design. The strong cut-off is its 75% quantile.

8. Evaluation against the proposal

All numbers in this section are from the run on the table to 13 October 2026 and are saved in `results_*.txt`, except the challenger (section 8.7), which ran on the table to 15 October. The live cards have since moved on to the report of 15 October (section 9).

8.1 Forecast value: does Dambulla add anything?

Colombo series	Horizon (report days)	Gain (points)	95% interval	Placebo gain	Items with gain	Verdict
Wholesale Pettah	1 / 3 / 5	+7.3 / +9.5 / +9.0	+6.4 to +11.3	+0.0 to +0.2	14 to 15 of 18	supported
Retail Pettah	1 / 3 / 5	+6.4 / +8.4 / +7.9	+5.5 to +10.3	0.0 to +0.1	14 to 16 of 19	supported
Retail Narahenpita	1 / 3 / 5	+7.5 / +12.2 / +12.0	+5.9 to +14.1	-0.3 to -0.1	10 to 17 of 19	supported; partly at 1 day

Gain is the share of squared forecast error removed, in percentage points, against the own-history model. The placebo swaps in another item's Dambulla price and shows nothing. The gain was similar in each test year, with one exception: retail Narahenpita at 1 day fell from +8.2 and +9.0 in 2024 and 2025 to +3.3 in 2026, and only 10 of 19 items gain, which is why that row is only "partly". In the fitted models roughly a sixth to a quarter of an unusual gap closed within 3 report days (gap coefficients 0.17 to 0.25).

8.2 Calibration: is the stated range honest?

Colombo series	Test cases	Inside the 80% range	By year (2024 / 2025 / 2026)	Average half-width
Wholesale Pettah	10,407	81.0%	80% / 82% / 81%	28.1%
Retail Pettah	10,739	81.5%	81% / 82% / 82%	23.3%
Retail Narahenpita	10,129	77.4%	79% / 76% / 77%	18.1%

The proposal's target was 75 to 85%, and all three series are inside it at 3 report days. Retail Narahenpita is the lowest, and its 2025 and 2026 values (76%, 77%) sit near the edge. The ranges are wide, which is the honest message: a 3-day change of a few percent is often smaller than the uncertainty.

The same check at the other two horizons (`results_model_h1.txt`, `results_model_h5.txt`):

Colombo series	1 report day: inside / half-width	5 report days: inside / half-width	5 report days by year (2024 / 2025 / 2026)
Wholesale Pettah	78.9% / 14.1%	81.6% / 39.0%	80% / 82% / 82%
Retail Pettah	79.9% / 11.5%	82.3% / 32.1%	82% / 83% / 82%
Retail Narahenpita	80.8% / 12.2%	76.1% / 22.7%	77% / 74% / 77%

All six overall figures are inside 75 to 85%. One single year is not: retail Narahenpita at 5 days in 2025 was 74%, which we report rather than hide. The website and dashboard show the 3-day range only.

8.3 Decision value: is it worth anything in rupees?

Colombo series	Dambulla model vs always buying today	Own-history model vs always buying today	Hindsight ceiling	Dambulla vs own history (95% interval)	Rs per 10 kg	Direction right: all / strongest 25%
Wholesale Pettah	-1.42%	+0.15%	-5.34%	-1.57 pts (-1.98 to -1.18)	-62	62% / 75%
Retail Pettah	-1.22%	-0.06%	-4.40%	-1.17 pts (-1.47 to -0.87)	-60	65% / 76%
Retail Narahenpita	-1.57%	-0.52%	-3.56%	-1.05 pts (-1.23 to -0.85)	-96	69% / 85%

Negative means cheaper. The simulated buyer buys once, today or 3 report days later, with the action threshold fixed on pre-2024 data. The signal captures roughly 27 to 44% of what perfect hindsight could save. This is a simulation, not observed savings: it assumes a purchase can be deferred at no cost, which small kitchens and perishable produce may not allow, and consecutive decisions overlap, so the month-resampled intervals are the honest guide.

8.4 Safe failure

`refusal_report.py` builds a small synthetic price table with exactly one fault each and records what the guard says. All 12 cases behave as specified (`results_refusal.txt`). These are tests of the logic on constructed data, not claims about any real report.

Proposal's test	Result	How it is tested
Stale data (over 3 working days)	Pass	Boundary test (3 allowed, 4 refused), weekend and holiday tests; model, dashboard and website tests check that every number disappears
A rejected or unreadable report	Pass	`test_refresh`: an unreadable new report is logged, not guessed, and the guard refuses
Newer report listed but not read	Pass	Guard, model, dashboard and website tests
A closed market	Pass	Pettah and Dambulla closures refuse; a fish-market closure correctly does not block vegetables
Missing Dambulla or Colombo price	Pass	Guard tests, missing and "n.a."
Too few observations	Pass	Guard test, and an empty table
Source unavailable	Pass	`test_refresh`, and the real robots.txt incident in section 6
A withheld card leaks no number through the website	Pass	`web/test_web.py` checks the cards and the price-history routes
The page never says "buy" or "sell"	Pass	The dashboard test scans every rendered view

8.5 Extraction accuracy

125 extracted prices match figures quoted in each report's own summary text across 2020 to 2026 samples, and today-versus-prior pairs across consecutive reports are checked (about 0.5% differ and are flagged). Of 1,495 reports in the supported layouts, 1,487 parse; the other 8 are listed with their reason.

8.6 The proposal's own "this is not working" tests

Proposal's failure condition	Triggered?	Note
Gain interval includes zero for most series, or placebo exceeds half the gain	No	All nine intervals are above zero; placebo is at most 0.3 in size
80% range covers under 70% or over 90%	No	76.1% to 82.3% across all three horizons
Flagged days no better than unflagged days	No	The strongest 25% of calls were right 75 to 85% of the time, against 62 to 69% across all calls
Buyers cannot say what a card means	No	All 3 completed Today; 1 needed help; median 4/5
Any refusal test fails	No	All 85 tests and all 12 refusal cases pass

8.7 What did not work, and what the results do not show

Jump persistence. We first tried to predict whether a one-day jump of 15% or more would last. About 57% persisted, and our models did no better than always guessing persistence (accuracy 56.6 to 56.7% against 57.2%, AUC 0.50 to 0.54). We set it aside and kept the scripts (`jump_check.py`, `model_check.py`).

Gradient boosting did not beat ridge. We built a challenger (`challenger_check.py`): scikit-learn's histogram gradient boosting on the same six features, the same split and the same scoring, with its settings fixed before the run and nothing tuned on the test years. The rule, also fixed in advance, was to adopt it only if at 3 report days its interval was above zero and its edge at least 0.5 points in every Colombo series. Result (`results_challenger.txt`, table to 15 October 2026):

Colombo series	Horizon	Ridge skill	Boosted skill	Edge (points)	95% interval
Wholesale Pettah	1 / 3 / 5	8.5 / 10.7 / 9.6%	7.9 / 10.0 / 9.2%	-0.6 / -0.6 / -0.4	[-1.2, +0.0] / [-1.4, +0.2] / [-1.3, +0.5]
Retail Pettah	1 / 3 / 5	7.4 / 9.3 / 8.3%	6.7 / 7.9 / 6.5%	-0.7 / -1.4 / -1.8	[-1.4, -0.0] / [-2.5, -0.4] / [-3.4, -0.4]
Retail Narahenpita	1 / 3 / 5	11.1 / 15.4 / 13.7%	11.3 / 14.6 / 13.2%	+0.2 / -0.8 / -0.5	[-0.6, +0.9] / [-1.4, -0.1] / [-1.2, +0.3]

The boosted model was worse than ridge in 7 of the 9 cases, and clearly worse (interval below zero) for retail Pettah and retail Narahenpita at 3 days. The decision is to keep the ridge model, which also has fewer moving parts and every number explainable. `python3 challenger_check.py --selftest` shows the rule works: on synthetic data where Colombo responds linearly it keeps ridge, and where it responds only above a threshold it adopts the boosted model.

Limits of the evidence. The effect is modest, rests on one train and test split, and covers only the 18 to 19 items with Dambulla wholesale prices. Horizons count report days, so a Monday spans a weekend. The test period (2024 onward) is calm. The signal may weaken in a disruption such as the 2022 crisis; that period sits in the training data, so we cannot say how the model would have done on it out of sample. Large expected moves can still be uncertain: some cards read "strong" while their 80% range crosses zero, and the page always shows the range.

9. Gaps and limitations

Gap	Effect and what we did
No buyer feedback yet	The central user claim, that a buyer would act on a card, is untested. Interviews are the next step.
Website not hosted	It runs locally with two commands (section 11). A free host is not chosen yet.
Full price table is not in the repository	The Central Bank says the data are public, but the full table is large and is generated locally. The repository holds the last 375 days of the series the website shows; a reviewer who wants the full history runs the downloader (about 6 hours at the required pace).
776 older reports unparsed (2017 to mid-2020)	The study starts on 28 July 2020. History before that cannot be tested.
8 reports in the supported layouts unparsed	4 scanned, 3 with an undated header, 1 duplicate. Listed in `failed_files.txt`; none guessed.
Many cards withheld	On the 15 October report (the latest in the live cards), 22 of 56 cards show numbers and 34 are withheld: 30 no clear signal, 3 no Dambulla price, 1 no Colombo price. This is by design, but it limits how much of the market is covered.
Holidays cover 2026 only	`data/holidays.csv` holds the 2026 public holidays. Bank holidays that are not public holidays, and dates after 2026, are missing, so good data can look stale around them. The guard refuses, which is the safe direction.
Daily refresh is manual	The user runs a script. A scheduled job needs a decision on where the price table lives.
Rounded prices, simulation assumptions	Cheap items move in steps; the rupee figures assume 10 kg and costless deferral.

10. What the user sees

The website has six pages. Welcome explains the project in plain language and shows the Sri Lanka emblem beside a prominent notice that this is an independent student service. Today is a market command centre: source health, summary counts, an expected-change and uncertainty chart, the strongest-signal list, filters, guarded CSV export and detailed cards. Explore compares paired Dambulla and Colombo history; Weekly report compares two complete five-report-day windows; Method reads evidence from the saved result files; and Help explains the labels. Every page carries CBSL attribution and non-endorsement wording. The figures below use real prices through 15 October 2026.

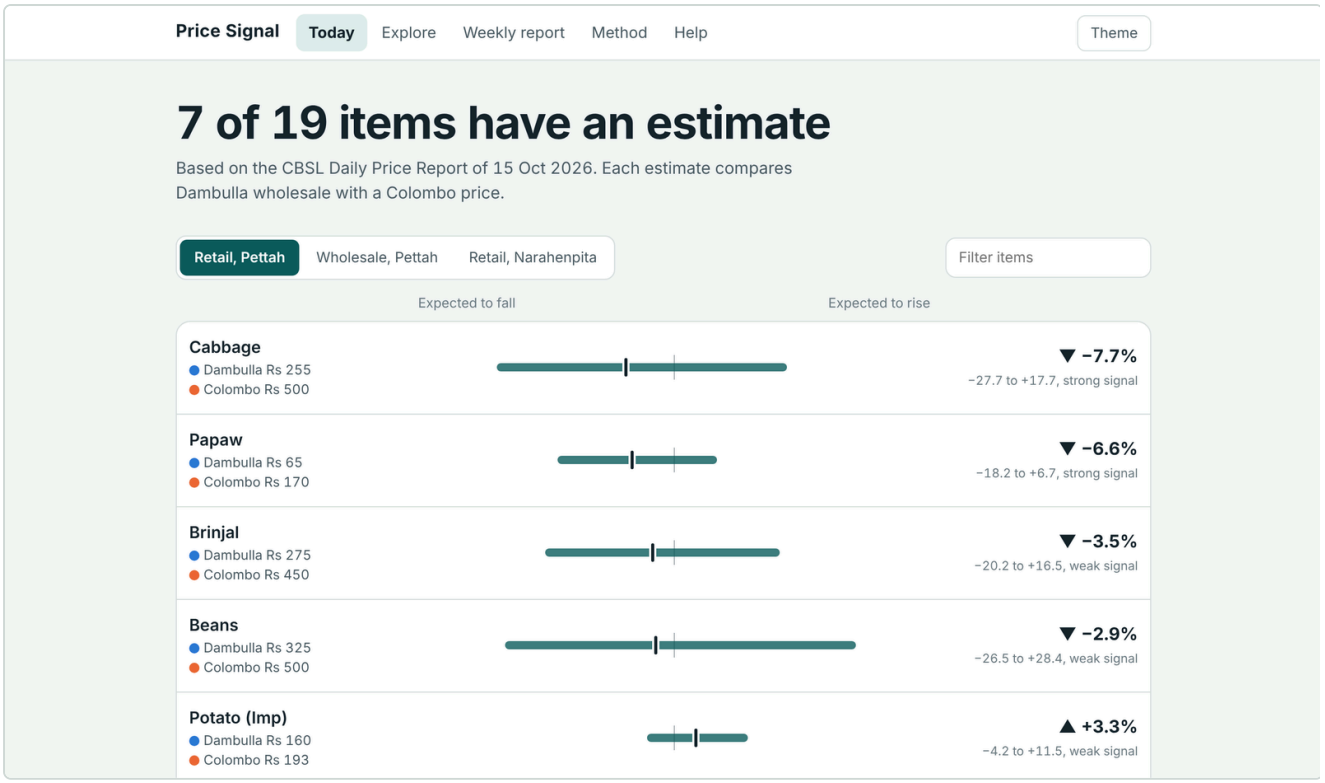


Figure 2. Today: report of 15 Oct 2026, one working day old. Arrows and signs carry the direction; the bar is the 80% range.

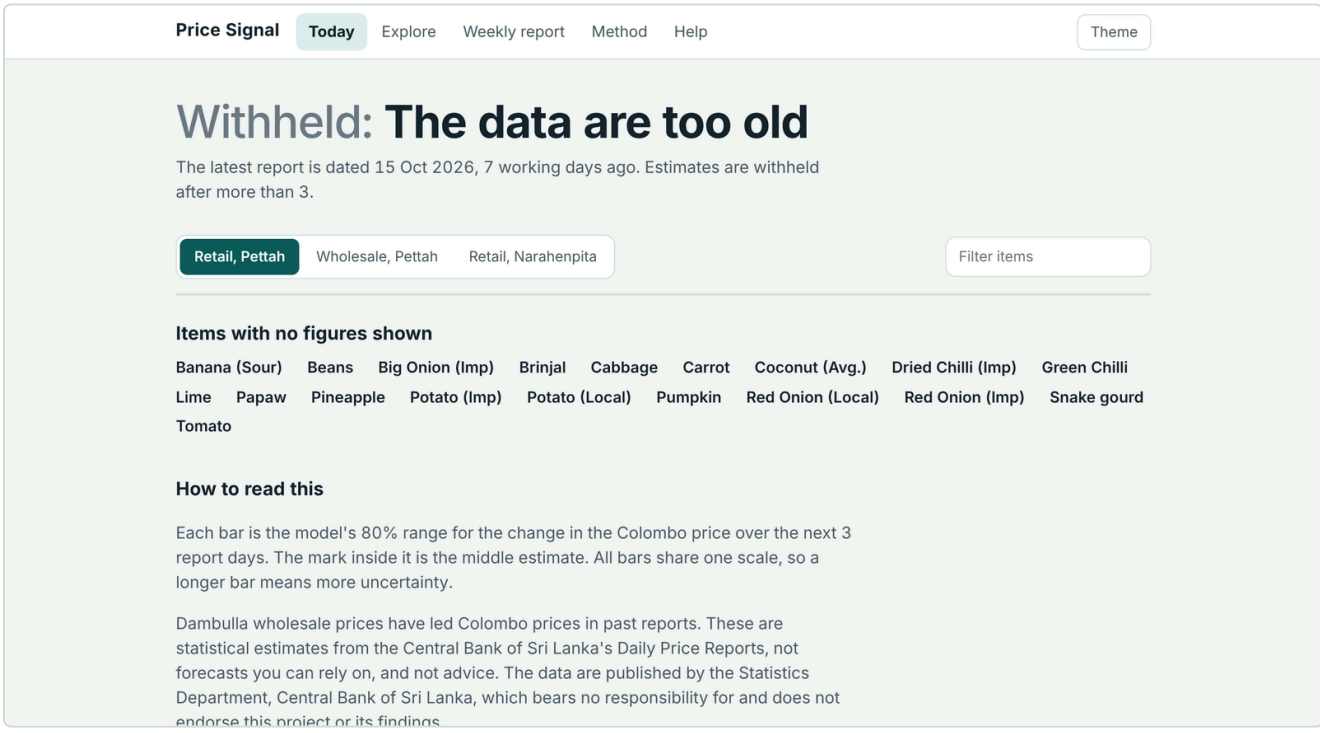


Figure 3. The same site viewed 8 working days later: every number is withheld and the reason is shown.

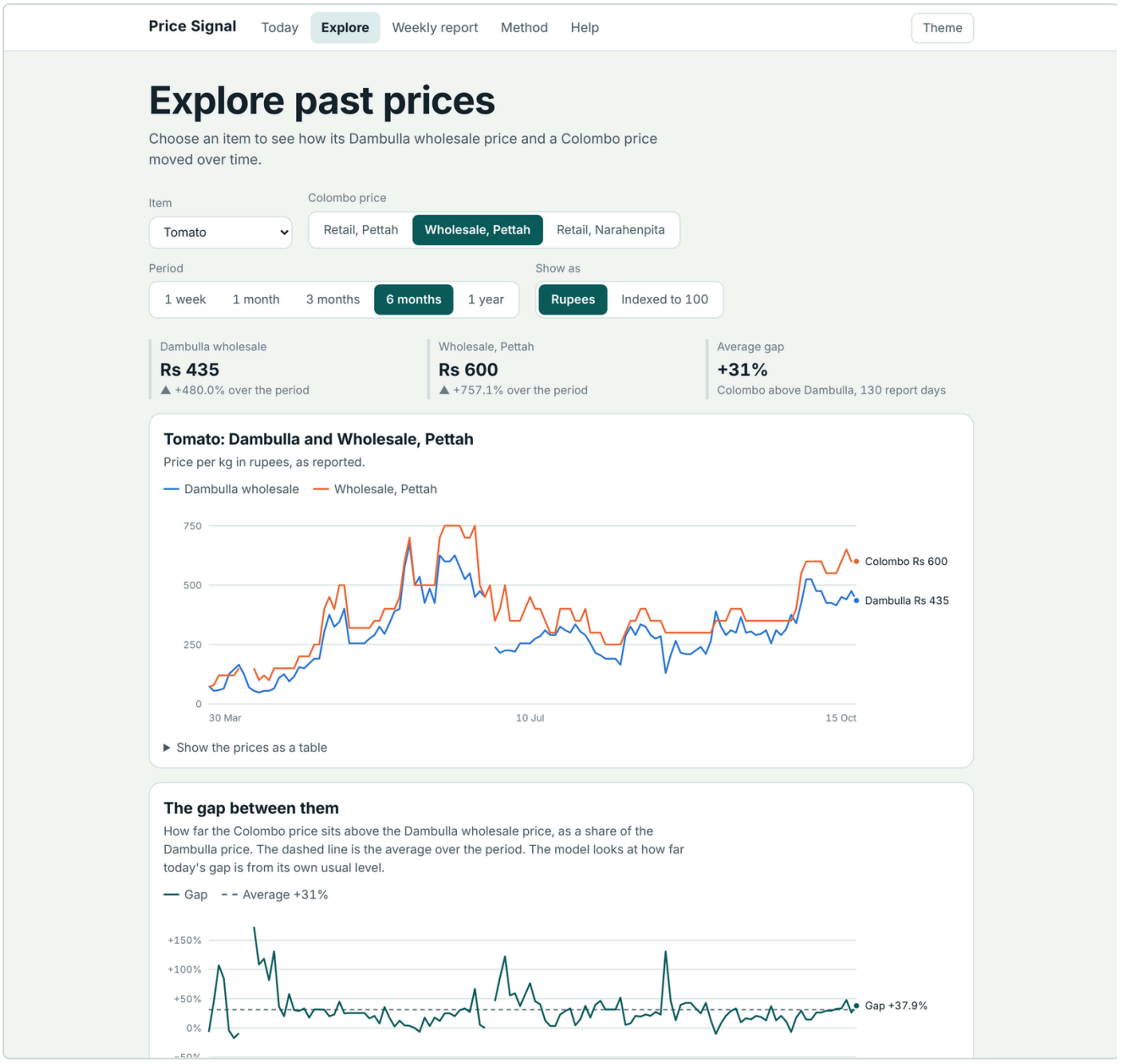


Figure 4. Explore: Tomato, Dambulla wholesale against wholesale Pettah, 6 months. A missing report day breaks the line instead of being filled in.

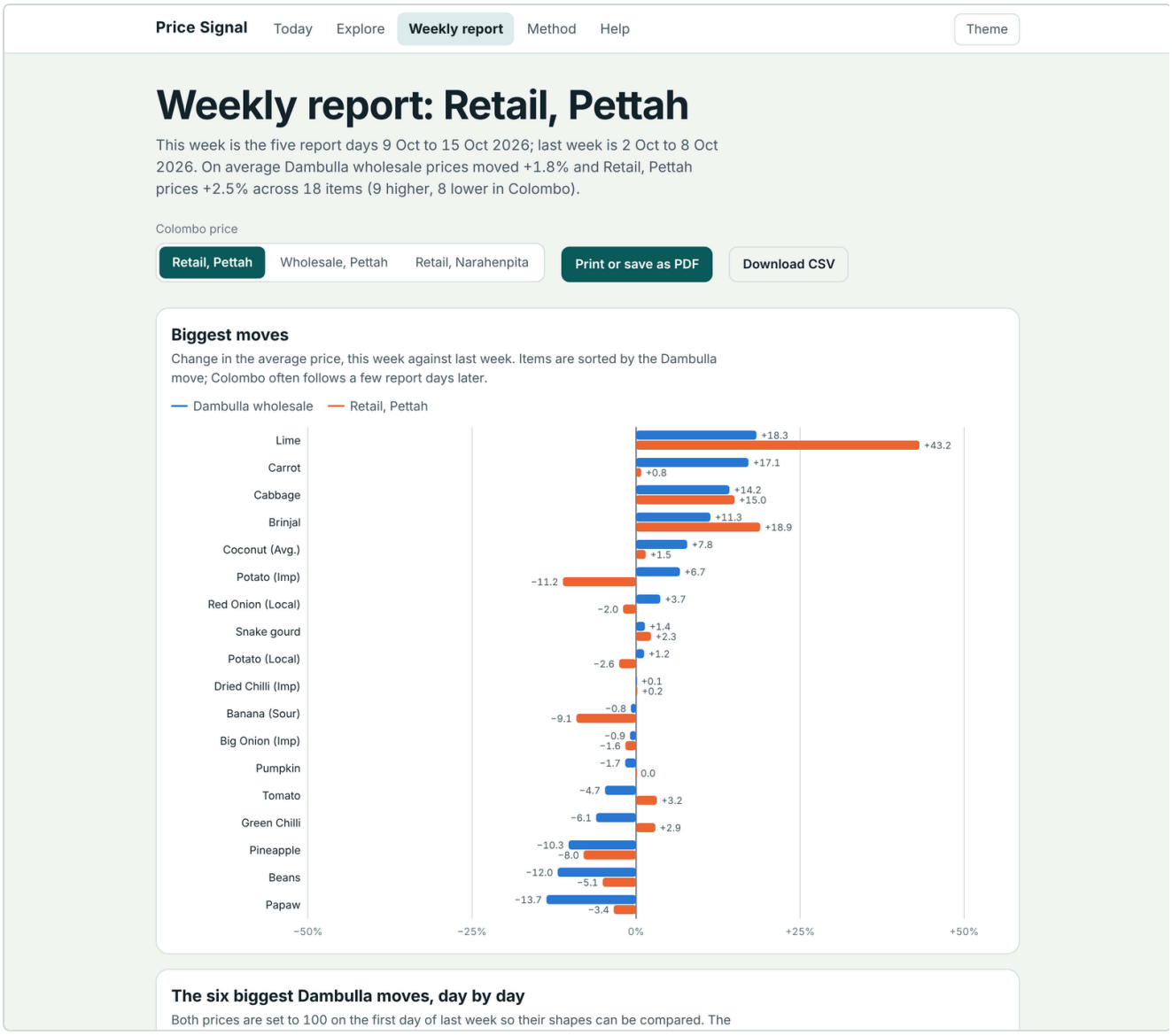


Figure 5. Weekly report for retail Pettah: this week (9 to 15 Oct) against last week (2 to 8 Oct).

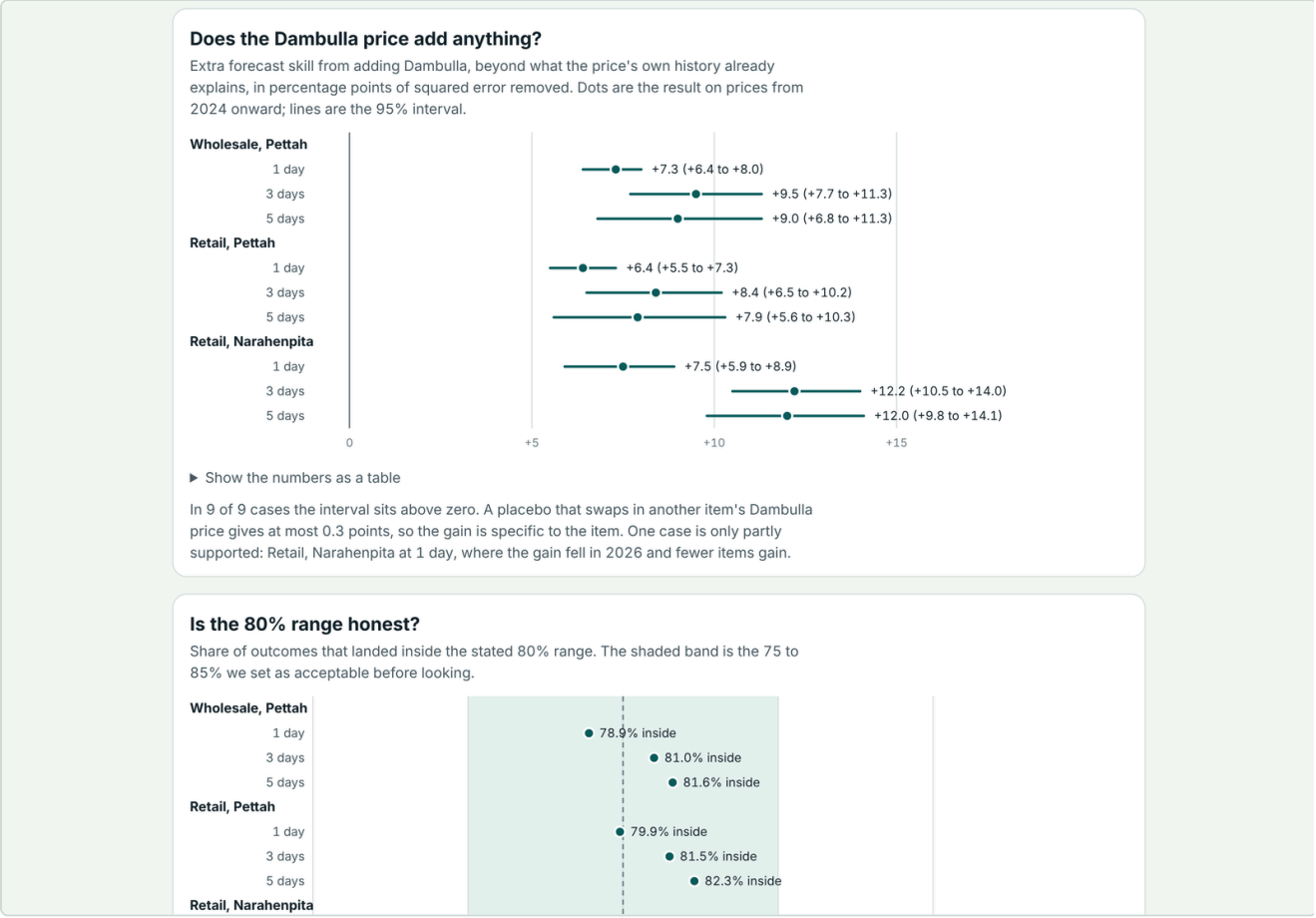


Figure 6. Method: the evidence charts, read from the repository's result files.

11. Reproducing the work

From a clean machine, in the repository folder:

```
pip install -r requirements.txt
python3 download_cbsl_reports.py        (about 6 hours at the 10-second delay; resumes)
python3 extract_prices.py data/raw prices_all.csv
python3 robust_check.py prices_all.csv
python3 leadlag_check.py prices_all.csv
python3 decision_backtest.py prices_all.csv
python3 model.py prices_all.csv --evaluate
python3 challenger_check.py prices_all.csv
python3 model.py prices_all.csv
python3 export_recent.py prices_all.csv
python3 web/build_db.py
python3 web/server.py                   (open http://127.0.0.1:5000; set PORT=5001 if the port
is busy)
python3 -m streamlit run app.py         (the second interface)
python3 -m pytest -q
python3 refusal_report.py
```

Each analysis prints the tables in this report. `model.py --selftest`, `robust_check.py --selftest` and `challenger_check.py --selftest` check the methods on synthetic worlds with known answers. A fresh clone runs the website from `cards.csv` and `data/recent_prices.csv`. The default repository suite now reports 85 passed and zero skipped. The stronger 125-value check against original CBSL PDFs remains an explicit local validation because those source reports are intentionally not redistributed.

12. AI assistance

Claude (Anthropic) and OpenAI Codex assisted with research, first drafts of code and documents, testing, review, safety hardening, the redacted permission evidence, the interface and this report update. The file-by-file account is in `AI_USE.md`. The project author ran and reviewed the work and remains responsible for every submitted claim and artefact.

13. Remaining submission actions

Three anonymous buyer sessions and the MIT licence are complete. The remaining submission actions are:

1. Finish the public Render service, verify `/healthz` and every application page, then replace the pending URL in the submission files.
2. Record the three-minute video from that exact deployed build and complete the signed-out browser check.
3. Re-export this report after inserting the verified URL so the PDF, README, summary and video all describe the same commit and data date.
4. Extend `data/holidays.csv` beyond 2026 before using the service in a later year.

14. Implementation update

Revision date: 18 October 2026 · This page records work completed after the 17 October report build.

14.1 Recent product changes

- A new Welcome page gives a non-technical introduction, uses the supplied Sri Lanka emblem and clearly states that the service is independent and not an official government website.
- Today was expanded into a command centre with source health, signal counts, an expected-change/uncertainty chart, a priority list, filters, safe CSV export and detailed cards.
- Technical labels were simplified across the interface while the Method page retains the evidence needed by reviewers.
- Three consented buyer sessions exposed confusion about the possible range, shared report days and blank weekly values; the site now explains each in plain language and shows the data date more prominently.

14.2 Final verification

Area	Verified state on 18 October 2026
Release	Functionality frozen at `v1.0.0-rc1`; later interface wording changes respond to buyer feedback and have regression coverage.
Automated checks	85 passed and zero skipped; 12 of 12 deliberate refusal cases pass; model, robustness and decision-policy self-tests pass.
Current build	Committed cards use the 15 October 2026 report: 22 of 56 cards show guarded estimates and 34 are withheld with reasons.
Buyer evidence	Three anonymous, consented sessions were completed on 16-17 October. Median understanding, trust and usefulness ratings were each 4/5.
Public evidence	The repository includes the redacted CBSL reply, MIT licence, saved results, AI-use disclosure, freeze record and anonymised buyer findings.

14.3 Buyer evaluation and response

A restaurant buyer participated on 16 October; a vegetable-shop buyer and a hotel/catering buyer participated on 17 October. The individual understanding scores were 4, 3 and 4; trust as one input was 4, 2 and 4; and usefulness was 5, 3 and 4. Two participants needed help or could not explain the possible range, all three needed help or did not complete the historical comparison, and two questioned blank weekly values. These findings led directly to the revised explanations. This is formative evidence from three people, not proof of adoption or realised savings, and the revised wording has not yet been retested.

14.4 Honest submission status

Buyer feedback, the licence and the current report update are complete. The public deployment URL and the three-minute video remain pending. This PDF deliberately does not guess a URL; after the Render service is verified, the URL must be inserted and the report re-exported, then the repository, website and video must be checked in a signed-out browser against the same commit and data date.